

H1. $S(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$, $G = \{ S(t) \mid t \in \mathbb{R} \}$.

under composition $*$ $\iff$ matrix multiplication, prove G is a group

$\forall t_1, t_2 \in \mathbb{R}$,

$$S(t_1) * S(t_2) = \begin{pmatrix} 1 & t_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & t_2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + t_1 \cdot 0 & 1 \cdot t_2 + t_1 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 0 & 0 \cdot t_2 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & t_1 + t_2 \\ 0 & 1 \end{pmatrix} = S(t_1 + t_2)$$

$\therefore t_1 + t_2 \in \mathbb{R}$ (addition is closed under $\mathbb{R}$).

$\therefore S(t_1) * S(t_2) \in G \Rightarrow *$ is closed under G .

$$\therefore S(t_2) * S(t_1) = \begin{pmatrix} 1 & t_1 + t_2 \\ 0 & 1 \end{pmatrix} = S(t_1) * S(t_2) \Rightarrow *$$
 is associative

$$\forall t \in \mathbb{R}, S(t) * S(0) = S(0) * S(t) = S(t+0) = S(t) \Rightarrow S(0) \text{ is identity.}$$

$$\forall t \in \mathbb{R}, S(t) * S(-t) = S(-t) * S(t) = S(t-t) = S(0) \Rightarrow \text{all } S(t) \in G \text{ has its reverse.}$$

$\therefore G$ is a group.